Charan Rongali

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.76
Dr.K.K.R's.Gowtham	Class XII (Central Board of Secondary Education)	2024	95.80%
Dr.K.K.R's.Gowtham	Class X (Central Board of Secondary Education)	2022	95.00%

PROJECTS

Analog-Digital Signal Acquisition & Reconstruction

- Designed an integrated ADC-DAC system to demonstrate high-fidelity analog signal sampling and reconstruction.
- Experimentally Validated the Nyquist Sampling Theorem by analyzing aliasing effects under varied sampling rates.
- Evaluated frequency response and signal integrity to ensure accurate reconstruction of analog waveforms.

Embedded Scientific Computing System (AVR)

- Developed an embedded scientific calculator on ATmega328P with LCD interfacing and keypad input in AVR-GCC.
- Implemented custom numerical algorithms for scientific functions without standard math libraries.
- Designed a recursive expression parser supporting operator precedence and multi-function evaluation.

Multi-Mode Digital Clock using AVR Microcontroller

- Built an interrupt-driven digital clock with clock, timer, and stopwatch modes using AVR-GCC on ATmega328P.
- Implemented BCD multiplexing and a 7447 decoder to drive six 7-segment displays with flicker-free output.
- Developed state-machine firmware with debounced buttons for smooth mode switching and time adjustment.

Pipelined Radix-2 FFT Processor (FPGA Implementation)

- Designed an 8-point Radix-2 FFT in Verilog with streaming complex inputs using a fully pipelined architecture.
- Verified functional correctness against MATLAB simulations and Intel FFT IP for bit-level accuracy.
- Optimized FPGA throughput and Fmax benchmarking against Intel IP approaching IP-level efficiency.

Pipelined FIR Filter Design & FPGA Implementation

- Designed and implemented a fixed-point FIR filter in Verilog using Direct Form, Optimized, Genvar-based architectures.
- Extended all three architectures with pipelining and validated correctness against Intel FIR IP on FPGA.
- Benchmarked resource utilization and throughput across pipelined and non-pipelined implementations.

Hotel Review Intelligence Engine - Python, Streamlit, NLP, Sentence-Transformers, FAISS

- Built an NLP pipeline analyzing 50,000+ hotel reviews to detect sentiment drift over time, using aspect scoring and trend regression.
- Developed a hybrid recommendation engine with a live simulator showing rank changes under time-decay weighting.
- Validated model reliability via correlation and weight-sensitivity testing instead of asserting accuracy.

Numerical Eigenvalue Solver (QR Decomposition)

- Implemented a Python-based eigenvalue solver using iterative QR Decomposition with Gram-Schmidt orthogonalization.
- Designed convergence criteria based on off-diagonal norm reduction ensuring high-precision eigenvalue extraction.
- Evaluated numerical stability and iterative convergence with configurable tolerance for scientific computing

applications.

SKILLS

EDA & Simulation Tools: Quartus Prime Lite, ModelSim, LTSpice, Altium Designer, MATLAB

Hardware Description Languages: Verilog

Programming Languages: C, C++, Python, HTML, CSS, JavaScript,MySQL

Developer Tools & Version Control: Git, GitHub, LaTeX

RELEVANT COURSES

Mathematics & Computing: Machine Learning, Matrix Theory, Linear Algebra, Probability & Random Processes, Vector Calculus, Differential Equations, Scientific Programming.

Computer Science: Introduction to Programming, Data Structures & Applications, Database Management Systems (DBMS-1).

Electrical & Electronics: Semiconductor Device Fundamentals, Analog Electronics, Digital Systems, Circuits & Network Analysis, Signal Processing, Control Systems, PCB Design and Layout.

POSITIONS OF RESPONSIBILITY

Web Head | Inventure (Yoga & Meditation Club, IITH)

- Built a fully responsive website from scratch to digitize club activities and registrations.
- Optimized UI/UX design to improve member engagement and streamline information access.

Web Developer | Abhyanth 2026 (Sports Fest, IITH)

- Developed a responsive, informative website to serve as the fest's primary digital hub.
- Modified site architecture to deliver real-time updates and event data to participants.

Sponsorship Head | SPIC MACAY

- Secured sponsorship for National Awardee artists via corporate outreach.
- Orchestrated financial partnerships to support large-scale cultural performances and events.

Hospitality & Transport Head | Elan & Vision (Techno-Cultural Fest)

- Overseeing requirements for artists and judges to ensure premium comfort and hospitality.
- Coordinating complex transport logistics and schedules for all event stakeholders.

EXTRACURRICULAR

NCC Cadet | National Cadet Corps

- Attended CATC-VII camp in Karimnagar, undergoing rigorous physical and discipline training.
- Performed a theatrical skit on overcoming language barriers to promote national unity and diversity.